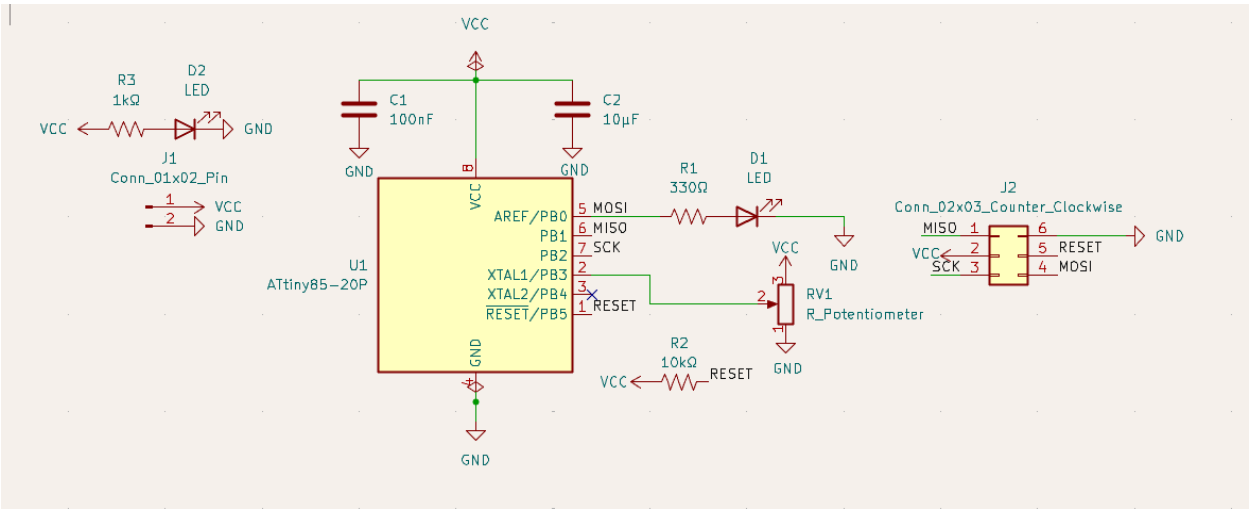
ATtiny85 Sensor Node PCB Writeup

7/27/2026

1. Overview

The sensor node is a small ATtiny85-based board designed to read an input from a potentiometer which outputs to a digital LED.. The board integrates a microcontroller (ATtiny85), analog sensing, digital I/O, and an in-system programming interface on a single 2-layer PCB.

2. Schematic



3. Component Selection

ATtiny85 (DIP-8 package): Chosen for its low pin count, 5V operation, and built-in ADC - everything needed for this project without the overhead of a larger microcontroller. The DIP-8 package was specifically selected over an SMD variant to make the chip hand-solderable (and socketable) without specialized reflow equipment.

100nF decoupling capacitor (across VCC/GND): Placed per datasheet recommendation for the ATtiny85, to filter high-frequency noise on the supply rail and keep the microcontroller's power input stable during switching events.

330Ω resistor (LED current limiting): Calculated value: with a 5V supply, a ~2V LED forward voltage drop, and a target drive current of 20mA

4. PCB Design Decisions

Decoupling capacitor placement close to VCC pin: The 100nF cap was placed as close as possible to the ATtiny85's VCC/GND pins to minimize trace inductance between the capacitor and the chip.

Through-hole components: Through-hole parts were chosen across the board to make the design hand-solderable.

2-layer board: A 2-layer stackup (top signal/component layer + bottom ground/routing layer) was sufficient for this design's low pin count and simple routing needs, while facilitating soldering.

Component Spacing: Components were generously spaced out to facilitate soldering

2nd LED: Added a second LED to confirm power from the 2-pin power supply was reaching the board, serving purely as a power indicator

5. PCB tests

1. **VCC voltage check**: With power applied, measure voltage at the VCC pin (pin 8) relative to GND with a multimeter; confirm it reads 5V.
2. **GND connectivity check**: Continuity-test GND pin (pin 4) against all other ground points on the board (decoupling cap ground leg, ISP header GND pin) to confirm unbroken ground plane/connection.
3. **LED functional test**: Drive PB0 high (via programmed firmware or direct ISP command) and confirm the LED turns on; drive it low and confirm it turns off, verifying the digital output path and current-limiting resistor are correctly wired.
4. **ADC functional test**: Sweep the potentiometer through its full range while reading the ADC channel connected to the wiper (via ISP readout); confirm the ADC value changes smoothly and monotonically with pot position.
5. **ISP header connectivity check**: Continuity-test each ISP header pin back to its corresponding ATtiny85 pin to confirm the programming header is correctly wired before attempting to program the chip.